Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Agentic LLM-Powered Digital Solutions for Healthcare, Research, and Industrial Automation**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio / Medical			
	Medium Category	Small Category	Medium Category	Small Category
	Open	Open		
Concept of Proposed Technology	CSEM brings both expertise in wearable smart devices and AI / GenAI solutions, and a digital platform for IoMT solutions [3][4] to the table to improve industrial and healthcare solutions to add value for patients, healthcare professionals and industrial solutions[1] [2]. The proposed technology is an **agentic system and LLM-powered digital platform** designed for healthcare, research, and industrial environments. It integrates autonomous decision-making with natural language understanding to: • Automate complex workflows such as triage, diagnostics support, and literature synthesis • Provide multimodal conversational interfaces for voice/text interaction • Interpret sensor data, clinical guidelines, and domain-specific datasets • Generate structured documentation and human-readable summaries • Deliver explainable, privacy-compliant, domain-tuned AI models The system combines **fine-tuned LLMs**, **agentic orchestration**, and **integration APIs** to support real-time, context-aware decision support across multiple sectors.			
Rationale for Joint R&D Project	A joint R&D project is essential because: **Healthcare and industrial partners require domain-specific adaptation**, which demands collaboration between AI research experts (CSEM) and Korean SMEs with market access and application-specific knowledge.			

	Finetuning LLMs on sensitive or proprietary datasets requires shared expertise in privacy, compliance, and secure data handling. **Deployment in real-world environments** (hospitals, factories, research labs) requires co-development of integration layers, validation protocols, and user-centric design. **Korean SMEs can accelerate commercialization**, localization, and regulatory alignment for the Korean market. Together, the partners can build a validated, scalable, and market-ready agentic LLM solution.
Key Words	• Agentic AI • Large Language Models (LLMs) • Digital Health / Clinical Decision Support • Multimodal Conversational Agents • Industrial Automation AI • Wearables • Vital Signs • Clinical studies • Data management

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 7
R&D Objectives	• Develop **agentic AI frameworks** capable of autonomous task execution in healthcare, research, and industrial settings • Finetune **domain specific- LLMs** ensuring accuracy, explainability, and privacy compliance • Build **multimodal conversational interfaces** (voice/text) for real-time- interaction • Develop **integration of APIs** for seamless connection to clinical systems, research databases, and industrial sensors • Validate prototype systems in sandboxed and real-world- environments			
R&D Target Specifications	• **LLM accuracy improvement**: ≥ 20% increase in domain specific- task accuracy (measured via benchmark datasets) • **Response latency**: < 1.5 seconds for standard inference tasks **Autonomous agent task completion rate**: ≥ 85% successful execution in defined workflows • **Documentation automation**: ≥ 70% reduction in manual documentation time • **Multilingual support**: At least 5 languages, including Korean and English • **Integration capability**: APIs compatible with HL7/FHIR (healthcare), OPC-UA (industry), and standard research data formats			
R&D Plan	**Phase 1 — Requirements & Architecture (Months 1–3)** • Joint workshops to define workflows and use cases • System architecture design (agentic framework + LLM modules) • Data collection and preprocessing **Phase 2 — Model Development & Fine-Tuning (Months 4–9)** • Finetuning LLMs on -domain specific- datasets • Development of autonomous agent modules • Privacy, safety, and explainability enhancements **Phase 3 — Prototype Development (Months 10–15)** • Build sandboxed testing environments • Develop conversational interfaces (voice/text) • Integrate APIs with partner systems **Phase 4 — Validation & Optimization (Months 16–21)** • Clinical, research, and industrial scenario testing • Performance benchmarking and optimization • User feedback cycles and UI/UX refinement **Phase 5 — Deployment Preparation (Months 22–24)** • Documentation and validation reports • Deployment support and technology transfer			

	• Commercialization roadmap

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Lead development of agentic AI frameworks and LLM fine-tuning • Provide technical expertise in AI architecture, multimodal interfaces, and integration • Build prototype systems and sandbox environments • Conduct validation studies and generate technical documentation • Ensure compliance with privacy, safety, and explainability standards • Support deployment and technology transfer
Expected R&R of Korean SME Partner	• Provide domain specific- requirements and datasets relevant to Korean market needs • Lead localization, UI/UX adaptation, and regulatory alignment • Conduct testing and validation in real-world- environments (clinics, labs, factories) • Develop integration layers with Korean digital health or industrial systems • Contribute to commercialization strategy and market deployment • Provide feedback for iterative improvement

5. Expected Outcomes/Impact

Expected Outcomes and Impact	• Development of a **validated agentic LLM platform** ready for deployment in healthcare, research, and industrial sectors • Significant **workflow automation**, reducing operational costs and improving efficiency • Enhanced **clinical decision support**, research productivity, and industrial monitoring accuracy • Strengthened **AI competitiveness** for both partners in Korean and global markets • Creation of **commercializable software modules**, APIs, and fine-tuned models • Contribution to Korea's national AI strategy through advanced, high impact- digital technologies

6. References

(1) Singhal, K., Azizi, S., Tu, T. *et al.* Large language models encode clinical knowledge. *Nature* 620, 172–180 (2023). https://doi.org/10.1038/s41586-023-06291-2

(2) F. Piccialli et al. "AgentAI: A comprehensive survey on autonomous agents in distributed AI for industry 4.0" Expert Systems with Applications, vol. 291, 2025, doi: https://doi.org/10.1016/j.eswa.2025.128404

(3) J. L. Aeberhard *et al.*, "Introducing Artificial Intelligence in Interpretation of Foetal Cardiotocography: Medical Dataset Curation and Preliminary Coding—An Interdisciplinary Project," *MPs*, vol. 7, no. 1, p. 5, Jan. 2024, doi: 10.3390/mps7010005

(4) C. Aguet *et al.*, "Sleep Staging in Patients with Suspected Sleep Apnea Using Wearables and Deep Learning," in *BMT 2025*, Muttenz, Switzerland: De Gruyter, Sep. 2025. doi: 10.1515/bmt-2025-1001.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization